COVID-19 Global Data Analysis

Exploratory Data Analysis and Regression Model Comparison

Project Report

Tools: Python, Pandas, NumPy, Matplotlib, Scikit-learn

1. Objective

Analyze COVID-19 case-related data, visualize patterns over time, and compare regression models that predict the next day's aggregated new-case count using recent daily values.

• Problem type: regression / time-ordered prediction.
• Models: Linear Regression, Polynomial Regression (degree 2), and Ridge Regression.

2. Dataset

The project CSV contains location-wise and date-wise fields. The analysis uses date and new_cases for daily aggregation; other available fields include total_cases, total_deaths, population, location, and continent.

Column	Meaning	Use
date	Date of observation	Time ordering
location	Country/location	Grouping context
new_cases	New cases recorded	Prediction target
total_cases	Cumulative cases	Dataset field
total_deaths	Cumulative deaths	Dataset field
population	Population	Dataset field
continent	Continent	Dataset field

3. Data Preparation and Method

• **Preparation:** Load the CSV with Pandas, convert dates, and sort records chronologically.

• **Aggregation:** Sum new_cases across locations for each date.

• **Outlier detection:** Apply the Interquartile Range (IQR) method to new_cases for individual records and daily aggregated totals. No IQR-defined outliers were detected in either check, so no values were removed through IQR-based outlier treatment.

• **Features:** Use the previous three daily totals as predictors for the next day's total.

• **Split:** Use the first 80% of time-ordered observations for training and the final 20% for testing.

• **Evaluation:** Fit each model and calculate MAE and R^2 on both training and testing data.

4. Models and Results

Three regression models were compared: Linear Regression, degree-2 Polynomial Regression, and Ridge Regression (alpha = 1.0). MAE measures average absolute prediction error (lower is better); R^2 measures fit relative to a mean-prediction baseline (higher is generally better).

Model	Test MAE	Test R^2
Linear Regression	43,178.64	0.0364
Polynomial Regression	67,843.71	-1.0048
Ridge Regression	43,178.64	0.0364

Result: Linear Regression and Ridge Regression tie for the lowest test MAE (43,178.64). Their test R^2 is 0.0364, indicating that they explain little of the variation in the test data. Polynomial Regression has a negative test R^2 (-1.0048), meaning it performs worse than a mean-prediction baseline on the test set.

5. Interpretation

Linear Regression and Ridge Regression have training and test R^2 scores close to zero, showing weak fit on both sets. This is consistent with a weak fit, though it does not prove the cause. Polynomial Regression has a stronger training score but much worse test performance, suggesting poor generalization and possible overfitting. These findings are based on one chronological split.

6. Conclusion

The project demonstrates data preparation, date-based aggregation, outlier checking, visualization, lag-feature engineering, and regression model evaluation. Linear Regression and Ridge Regression performed best by test MAE, but predictive performance was weak. The results show the importance of evaluating models on unseen data and comparing them with a simple baseline before relying on forecasts.

7. How to Run

```
pip install pandas numpy matplotlib scikit-learn
python3 project.py
python3 linear_regression.py
python3 compare_regression.py
```

Repository: https://github.com/Harshita-madaan/Covid-19-Mock-Analysis